Parabola

Mind map

Conic basics

- $PS = e \cdot PM$
- $\Delta \neq 0,\ h^2 = ab$

Standard $y^2 = 4ax$

- $S(a, 0)$, $x = -a$
- LR $4a$, ends $(a, \pm 2a)$
- $PS = x_1 + a$

Other and shifted

- $y^2 = -4ax$, $x^2 = \pm 4ay$
- $(y - k)^2 = 4a(x - h)$

Oblique axis

- $PS^2 = PM^2$
- $L_1^2 = 4aL_2$ (normalised)

Parametric point

- $(at^2, 2at)$
- slope $\frac{2}{t_1 + t_2}$

Focal chord

- $t_1 t_2 = -1$
- $4a\,\mathrm{cosec}^2\alpha$
- $\frac{1}{SP} + \frac{1}{SQ} = \frac{1}{a}$

Chords

- $2x - (t_1 + t_2)y + 2at_1t_2 = 0$
- 90° at vertex: $(4a, 0)$

Position of point

- $S_1 < 0$ inside
- $S_1 > 0$ outside

Tangent

- $ty = x + at^2$
- $y = mx + \frac{a}{m}$
- meet $(at_1t_2, a(t_1 + t_2))$

Director circle

- directrix $x + a = 0$
- $hm^2 - km + a = 0$

Normal

- $y = -tx + 2at + at^3$
- $t_2 = -t_1 - \frac{2}{t_1}$
- $t_1 t_2 = 2$ (meet on curve)

Normals from a point

- $am^3 + (2a - h)m + k = 0$
- $\sum t = 0$, centroid on axis

Transformation

- $a \to -a$; swap x, y
- $x \to x - h$, $y \to y - k$

Common tangents

- $y = mx + \frac{a}{m}$ + other condition
- shortest distance: common normal

Chord of contact, pole

- $T = 0$, $T = S_1$, $SS_1 = T^2$
- pole $\left(\frac{n}{l}, -\frac{2am}{l}\right)$
- diameter $y = \frac{2a}{m}$

Highlights

- 90° at focus; foot on $x = 0$
- $ST = SG = SP$; reflection
- area ratio $2 : 1$